

UNIVERSITY OF ENGINEERING AND TECHNOLOGY,
LAHORE

ENTRANCE TEST – 2010
For F.Sc and Non-F.Sc. Students
Time Allowed: 100 Minutes
Total MCQs: 100

Instructions:

- (i) Read the instruction on the MCQ Response Form carefully.
- (ii) Choose the single best answer for each question.
- (iii) Candidates are strictly prohibited from giving any identification mark except Roll No. & Signature in the specific columns only.

COMPULSARY QUESTION FOR IDENTIFICATION

Q-ID What is the color of your question Paper?

- A) BLUE
- B) GREEN
- C) RED
- D) YELLOW

Ans: Color of your question Paper is green. Fill the corresponding to letter 'B' Against 'ID' in your MCQ (Exactly as shown in the Diagram).

	A	B	C	D
-ID	☐	☒	☐	☐
-1	☐	☐	☐	☐
-2	☐	☐	☐	☐
-3	☐	☐	☐	☐

circle
response form

PHYSICS

- 1. One light year is equal to:
A) $9 \times 10^{12}m$
B) 6×10^{12}
C) $9.5 \times 10^{13}m$
D) None
- 2. Reverse process of vector addition?
A) Subtraction
B) Division
C) Multiplication
D) None
- 3. When a ball moves from and reaches to its maximum height in 3 seconds, what is its initial velocity?
A) 40m/sec
B) 29.4m/sec
C) 35m/sec
D) 25m/sec
- 4. Rocket propulsion is according to?
A) 3rd law of motion
B) 1st
C) 2nd
D) None
- 5. 1 Cal = _____ j
A) 418
B) 400
C) 10
D) 4.18
- 6. If a body of 2kg moves vertically 2m then work done is:
A) -39.2 j
B) 20 j
C) 45 j
D) 35 j
- 7. if the radius of circle is doubled, then centripetal force becomes:
A) Doubled
B) Remains same
C) Half
D) Reduces four times
- 8. Bernoulli equation is according to _____ law.
A) Stock's law
B) Einstiens equation
C) Newton's law
D) None
- 9. The time period of a second pendulum from its extreme to mean position:
A) T
B) T/4
C) T/2
D) 2 sec
- 10. At what temp, the velocity of sound is doubled than as that of at 10⁰c:
A) 859°C
B) 560°C
C) 760°C
D) 500°C
- 11. The range of visible wavelength spectrum is:
A) 400-750
B) 100-400
C) 400-1000
D) 300-600
- 12. The range of wave no. of light is $5 \times 10^6m^{-1}$. Then its wavelength is:
A) 2×10^7m
B) $2 \times 10^{-7}m$
C) $5 \times 10^{-8}m$
D) None
- 13. Least distance of distinct vision for human eye is:
A) 50 cm
B) 25 cm
C) 30 cm
D) 20 cm

14. If the the distance is doubled b/w two charges, then coulombs force becomes:
 A) Doubled B) Half
 C) Reduced 4 times D) Remains same
15. Two resistance of 6Ω each are in paralalled and third of 3Ω is connected in series.
 A) $10/3$ B) 15
 C) 9 D) 6
16. If a magnet is non-uniform magnetic field, it will experience:
 A) Attraction B) Repulsion
 C) Neutral D) None
17. $\text{Emf} = VBL \sin \theta$ (related to that):
 A) Volts B) Ohm
 C) Newton's D) Joules
18. The force F experienced by a conductor of length L carrying current when placed in a uniform magnetic field B is given by the relation.
 A) $F = B/iL$ B) $F = IL/B$
 C) $F = BiL$ D) $f = L \times B/I$
19. High voltage is obtai9ned form?
 A) Plasma generators B) Battery
 C) Energy prob D) None
20. Unit of longitudinal strain is:
 A) m B) Unit less
 C) $1/m$ D) None of these
21. A completely filled band is called:
 A) Forbidden band B) Conduction band
 C) Both D) None
22. Which of the following is used for detection of light?
 A) Reverse biased photodiode B) Forward biased photodiode
 C) Both D) None
23. In ${}_{92}\text{U}^{235}$ the neutron to nucleon ratio is:
 A) $143/235$ B) $145/235$
 C) $235/143$ D) $92/235$
24. If intensity and frequency of light is doubled, then emission of electrons is:
 A) Doubled B) Increases four times
 C) Half D) Same
25. When effective value of current is 10. What is its peak value?
 A) 12 B) 10
 C) 13 D) 14.5
26. In diodes, the graph is drawn b/w?
 A) Voltage & current B) Voltage and intensity
 C) Intensity and current D) None
27. Speed of earth around the sun is:
 A) 29600 m/sec B) 29000 m/sec
 C) 30000 m/sec D) None of above
28. In $v=wr$, w is called:
 A) Tangential velocity B) Radius of circle
 C) Rotational velocity D) All
29. A small parallel resistance (shunt) used to convert a galvanometer into an ammeter of suitable range is given by:
 A) $R_s = \frac{l_g}{l-l_g} R_g$ B) $R_s = \frac{l_g-l}{l_g}$
 C) $R_s = \frac{l_g}{l-l_g}$ D) None
30. A bullet is fired with an angle of 30° with the velocity of 196 ms^{-1} . For how much time the bullet will remain in space?
 A) 10 s B) 30 s
 C) 20 s D) 40 s

MATHEMATICS

31. Which of the following does not have closure property?
 A) $[0, 1]$ B) $[1 - 1]$
 C) $[0, 0]$ D) None
32. A non-negative rational number is group under:
 A) Addition B) Multiplication
 C) Subtraction D) Both A and C
33. A set with no element is called:
 A) Empty set B) Unit set
 C) Identity set D) None
34. A matrix with all entry zero is called:
 A) Null matrix B) Unit matrix
 C) Identity matrix D) None
35. If degree of polynomial in numerator is less than denominator then fraction is called?
 A) Proper fraction B) Improper fraction
 C) Both D) None
36. If "a" is 1st term and "r" is common ratio, then $(n + 1)$ th term is?
 A) ar^n B) ar
 C) a D) None
37. Binomial theorem is used to sum up the following series:
 A) Divergent B) Convergent
 C) Any D) None
38. $\sin(180 + \theta)$:
 A) $-\sin \theta$ B) $-\cos \theta$
 C) Both D) None
39. $\frac{\cos 11^\circ + \sin 11^\circ}{\cos 11^\circ - \sin 11^\circ} = ?$:
 A) $\tan 56^\circ$ B) $\sin 56^\circ$
 C) $\cos 56^\circ$ D) None
40. The solution of $1 + \cos x = 0$ is:
 A) $\pi + 2k\pi, K \in \mathbb{Z}$ B) $\pi - 2k\pi$
 C) $2k\pi$ D) None
41. $\int x \cos x dx = ?$
 A) $x \sin x - \cos x + c$ B) $x \sin x - \cos x - c$
 C) $x \cos x - \sin x + c$ D) None
42. $\lim_{\theta \rightarrow 0} \frac{\sin^2 \theta}{\theta} = ?$
 A) 0 B) 1
 C) -1 D) 2
43. If $x = a \cos \theta, y = a \sin \theta$ then $\frac{dy}{dz} = ?$
 A) $-\cot \theta$ B) $\cos \theta$
 C) $\tan \theta$ D) None
44. If $y = \sqrt{a^2 + x^2}$, then $\frac{dy}{dx} = ?$
 A) $\left(\frac{x}{\sqrt{a^2 + x^2}} \right)$ B) $\frac{\sqrt{a^2 + x^2}}{x}$
 C) $\left(x\sqrt{a^2 - x^2} \right)$ D) None
45. If "k" is constant then $\int k, f(x) dx = ?$
 A) $k, \int f(x) dx$ B) $k \int (x)$
 C) $F(x)$ D) None
46. If $\frac{dy}{dx} = 2x$, then the equation becomes:
 A) $y = x^2 - 2x - 3$ B) $y = x^2 - 2x + 3$
 C) $Yx^2 + 2x - 3$ D) None
47. If eccentricity $e > 1$, then the figure is:
 A) Ellipse B) Parabola
 C) Circle D) Hyperbola

48. If $y^2 = 3x$, then the equation of rectum is”
 A) $x = \frac{3}{4}$ B) $Y = \frac{4}{3}$
 C) $Y = \frac{1}{3}$ D) $Y = \frac{5}{7}$
49. The probability of numbers greater than ‘4’ by throwing a dice is:
 A) $\frac{1}{3}$ B) 3
 C) $\frac{4}{3}$ D) $\frac{2}{3}$
50. $\left(\frac{1-\sqrt{3}i}{2}\right)^4$:
 A) $\frac{-1+\sqrt{3}i}{2}$ B) $\frac{-1-\sqrt{3}i}{2}$
 C) $\left(\frac{-1+\sqrt{3}i}{2}\right)^2$ D) None
51. When $f(x) = \sin x$ and $x = 0$, then slope will be:
 A) 1 B) -1
 C) 0 D) 2
52. $R = ?$
 A) $\frac{abc}{4\Delta}$ B) abc
 C) $\frac{abc}{\Delta}$ D) None
53. $\sinh^2 x + \cosh^2 x = ?$:
 A) $\cosh 2x$ B) $\cosh x$
 C) $\sinh 2x$ D) None
54. $\frac{1 - \cos \alpha}{\sin \alpha} = ?$:
 A) $\tan \alpha / 2$ B) $\cot \alpha / 2$
 C) $\sin \alpha / 2$ D) None
55. If $ax^2 + 2hxy + by^2 = 0$, their lines are orthogonal when:
 A) $a + b = 0$ B) $a - b = 0$
 C) $a = b$ D) None
56. The distance of a point (10,- 10) from origin is:
 A) $10\sqrt{2}$ B) 10
 C) 15 D) None
57. The A.M between 1000 and 4 is?
 A) 502 B) 500
 C) 402 D) 602
58. Area of Δ :in terms of sides is equal to:
 A) $S(s-a)(s-b)(s-c)$ B) $\sqrt{S(s-a)(s-b)(s-c)}$
 C) $S(s+a)(s+b)(s+c)$ D) None
59. If $\sin \theta < 0$ and $\cos \theta > 0$ then in which quadrant terminal sides of the angle lies?
 A) 1st B) 2nd
 C) 3rd D) 4th
60. A dice is rolled, what is the probability of getting a number greater than 4?
 A) $\frac{1}{4}$ B) $\frac{1}{3}$
 C) $\frac{1}{2}$ D) $\frac{1}{6}$

CHEMISTRY

61. The roots of combined equations of straight line $ax^2 + 2hxy + by^2 = 0$ are real and coincident if:
 A) $h^2 - ab > 0$ B) $h^2 - ab < 0$
 C) $h^2 = ab$ D) $h^2 = -ab$
62. Which is universal solvent?

- A) Water
C) Ether
63. **Molarity of heavy water:**
A) 55.5 mol.dm³
C) 40.5
64. **Diamond is non-conductor due to:**
A) Free electrons present
C) Absence of free electron
65. **Which of the following is true?**
A) Na⁺ is smaller than Na
C) Na⁺ is greater than Na
66. **Which of the following is not alkali?**
A) Ra
C) Na
67. **The comparative rate at which solute moves in paper chromatography, depends upon R_f value of:**
A) Solute
B) Solvent
68. **In primary alkyl halides, hydrogen is bonded to that carbon which is attached to carbon atoms?**
A) 1
C) 2
69. **Which of the following is least reactive?**
A) H₂O
C) Ether
70. **Which of the following is the formula of methylene chloride?**
A) CH₂Cl₂
C) CH₃Cl
71. **Which of the following has net dipole moment?**
A) CO
C) H₂O
72. **The maximum value of O₂ is at:**
A) HTP
C) 127°C, 1atm
73. **The order of rate of diffusion is:**
A) SO₂>CO₂>Br₂>NH₃
C) Br₂>SO₂>CO₂>NH₃
74. **Oxidation of NO in air gives:**
A) N₂H₄
C) NH₃
75. **Which of the following is used for disinfection of water?**
A) Cl₂
C) Br₂
76. **The planar shape is made in which type of hybridization?**
A) sp³
C) sp¹
77. **In which orbital electrons go after 6d?**
A) 7s
C) 7p
78. **At which point heat is absorbed without change in phase?**
A) Melting point
C) Boiling point
79. **Which gives +3 charge?**
A) Al
C) Na
80. **The pH of 10⁻³ moles of HNO₃ is?**
A) 2.7
C) 3
- B) HCl
D) None
- B) 60.5
D) 50.5
- B) Giant structure
D) None
- B) Cl⁻ is less than Cl
D) none
- B) K
D) All
- B) Solution
D) None
- B) 3
D) None
- B) Alcohol
D) H₂SO₄
- B) CHCl₃
D) CCl₄
- B) NH₃
D) All of these
- B) STP
D) None
- B) Br₂<SO₂<CO₂<NH₃
D) SO₂<CO₂<Br₂<NH₃
- B) NH₄
D) None
- B) F₂
D) I₂
- B) sp²
D) None
- B) 8s
D) 6s
- B) Both
D) None
- B) Ca
D) None
- B) 2.0
D) 1.5
- B) Marble
D) Marine shell
- B) Polyethylene
D) None
81. **Which is not a calcareous material?**
A) Clay
C) Lime
82. **Which is synthetic polymer?**
A) Polystyrene
C) PVC
83. **Which is not present abundantly in earth crust?**

- A) Na
C) Ba
- B) Al
D) Ca
84. Which catalyst is used to convert carboxylic acid into alcohol?
A) NaBH_4
C) LiAlH_4
B) H_2SO_4
D) None
85. Methylated Alcohol contains:
A) 95% alcohol
C) 80 % alcohol
B) 90 % alcohol
D) None
86. A single chloride free radical can destroy how many ozone molecules:
A) 100
C) 100000
B) 10000
D) 10
87. How many zones through which the charge passes in rotary kiln?
A) 4
C) 3
B) 2
D) 5
88. Which compound shows H-bonding?
A) C_2H_4
C) $\text{C}_2\text{H}_2\text{Cl}$
B) $\text{CH}_3-\text{O}-\text{CH}_3$
D) $\text{C}_2\text{H}_5\text{OH}$
89. London dispersion forces exist in:
A) In bromine water
C) In HCL
B) In molecule of neon gas
D) In molecules of CH_3Cl
90. The reactivity order of alkyl halides for particular alkyl group is:
A) $\text{F} > \text{Cl} > \text{Br} > \text{I}$
C) $\text{Cl} > \text{Br} > \text{F} > \text{I}$
B) $\text{I} > \text{Br} > \text{Cl} > \text{F}$
D) $\text{Br} > \text{I} > \text{Cl} > \text{F}$

ENGLISH

Spot the error in the following sentences.

91. This apartmentbe perfect, if it were a little larger.
 A B C D
92. He borrowed the money so he could finishhis education.
 A B C D
93. The duties of the new secretary are to answer the telephone, to type letters, and book
 A B C
Keeping:
 D
94. Daina would rather that her husband doesn't work sohard.
 A B C D

Pick the correct one:

95. A)She drenched herself in perfume
 B)She drenched herself with perfume
 C)She drenched herself into perfume.
 D)She drenched herself in perfume.
96. A)A day later he reached his first glimpse of Lahore
 B)A day later he took his first glimpse of Lahore.
 C)A day later he found his first glimpse of Lahore
 D)A day later he caught his first glimpse of Lahore.
97. A)This has a bad impact on economy.
 B)This has a bad impact to the economy.
 C)This has a bad impact for the economy.
 D)This has a bad impact over the economy.
98. A)He saved him from dying of thirst
 B)He saved him for dying of thirst
 C)He saved him from dying from thirst
 D)He saved him for dying from thirst
99. A)They try to pacify him for kindness and affection
 B)They try to pacify him in kindness and affection.
 C)They try to pacify him by kindness and affection.
 D)They try to pacify him with kindness and affection.
100. A)Why did you disagree with me?
 B)Why did you disagree to me?
 C)Why did you disagree on me?
 D)Why did you disagree by me?